Synchronous Generator

Formula sheet for the $xy0$ machine model

Notation. Subscripts S, E, DX, and DY denote the stator, excitation winding, and the damper windings on the x- and y-axes. The dot denotes a time derivative, e.g. $\dot{\psi} = \mathrm{d}\psi/\,\mathrm{d}t$. The symbols x and y correspond to the rotating machine axes used in the source model.

Clarke transformation: three-phase to $xy0$

$$u_0 = \frac{u_1 + u_2 + u_3}{3} = R_0 i_0 + L_0 \dot{i}_0, \tag{1}$$

$$u_{SXS} = \frac{2u_1 - (u_2 + u_3)}{3}, \tag{2}$$

$$u_{SYS} = \frac{u_2 - u_3}{\sqrt{3}}. \tag{3}$$

The stationary components are rotated through the electrical angle α:

$$u_{SX} = u_{SXS}\cos\alpha + u_{SYS}\sin\alpha, \tag{4}$$

$$u_{SY} = u_{SYS}\cos\alpha - u_{SXS}\sin\alpha. \tag{5}$$

Electrical equations in the rotating reference frame

$$u_{SX} = R_S i_{SX} + \dot{\psi}_{SX} - p\omega\psi_{SY}, \tag{6}$$

$$g u_E = g^2 \frac{2}{3} R_E i_E + \dot{\psi}_E, \tag{7}$$

$$0 = R_{DX} i_{DX} + \dot{\psi}_{DX}, \tag{8}$$

$$u_{SY} = R_S i_{SY} + \dot{\psi}_{SY} + p\omega\psi_{SX}, \tag{9}$$

$$0 = R_{DY} i_{DY} + \dot{\psi}_{DY}. \tag{10}$$

Flux linkages and current components

The common magnetizing flux linkages are

$$\psi_{hX} = \psi_{SX} - L_S i_{SX} = \psi_{DX} - L_{DX} i_{DX} = \psi_E - L_E \frac{i_E}{g}, \tag{11}$$

$$\psi_{hY} = \psi_{SY} - L_S i_{SY} = \psi_{DY} - L_{DY} i_{DY}. \tag{12}$$

The associated magnetizing currents are

$$i_{\mu X} = f(\psi_{hX}) = i_{SX} + i_{DX} + \frac{i_E}{g}, \tag{13}$$

$$i_{\mu Y} = f(\psi_{hY}) = i_{SY} + i_{DY}. \tag{14}$$

For the nonlinear magnetization characteristic shown in the source model,

$$i_\mu(\psi) = i_{\mu N} \left[0.9 \frac{\psi}{\psi_N} + 0.1 \left(\frac{\psi}{\psi_N} \right)^7 \right]. \tag{15}$$

Mechanical equations

$$M = M_{RBG} - M_i, \tag{16}$$

$$M_i = \frac{3}{2} p \left(i_{SY} \psi_{hX} - i_{SX} \psi_{hY} \right), \tag{17}$$

$$\alpha = p \int \omega \, \mathrm{d}t, \tag{18}$$

$$M_{RBG} = M_{\text{Reibung}} \, \text{sign}(n). \tag{19}$$

Here, p is the number of pole pairs and n is the mechanical speed.

Inverse transformation: $xy0$ to three-phase

First rotate from the machine frame back to stationary coordinates:

$$i_{SXS} = i_{SX} \cos\alpha - i_{SY} \sin\alpha, \tag{20}$$

$$i_{SYS} = i_{SX} \sin\alpha + i_{SY} \cos\alpha. \tag{21}$$

Then recover the phase currents:

$$i_1 = i_{SXS} + i_0, \tag{22}$$

$$i_2 = \frac{-i_{SXS} + \sqrt{3}\, i_{SYS}}{2} + i_0, \tag{23}$$

$$i_3 = \frac{-i_{SXS} - \sqrt{3}\, i_{SYS}}{2} + i_0. \tag{24}$$

Default model simplifications (if not otherwise supported by the parameter line):

$$\frac{i_E}{g} = i'_E, \qquad L_E = L_S, \quad L_0 = L_S, \quad L_{DX} = L_S, \quad L_{DY} = L_S,$$

$$R_0 = R_S, \qquad R_{DX} = R_D, \qquad R_{DY} = R_D.$$

Constant machine parameters

$$U_N = \frac{U_{N,\text{verkettet}}}{\sqrt{3}}, \qquad Z_N = \frac{U_N}{I_N}, \qquad \omega_{EL} = 2\pi f_N, \tag{25}$$

$$R_S = Z_N\, r_{cu}, \qquad L_S = \frac{\text{Streuung}\, Z_N}{K_C \omega_{EL}}, \qquad \Psi_N = \frac{\sqrt{2}\, U_N}{\omega_{EL}}, \tag{26}$$

$$g = \frac{I_{E0}(1 - \text{Streuung})}{K_C \sqrt{2}\, I_N}, \qquad R_D = \frac{R_S}{\text{Daempferfaktor}}, \qquad I_{MNY} = \frac{I_{MNX}}{\text{Schenkelpolfaktor}}, \tag{27}$$

$$I_{MNX} = \frac{I_{E0}}{g}, \qquad M_N = \frac{pP_{EL}(1 - r_{bg} - r_{cu})}{\omega_{EL}}. \tag{28}$$

Parameter definitions

Symbol	Meaning
Streuung	Leakage factor
Daempferfaktor	Damping factor
Schenkelpolfaktor	Salient-pole factor
r_{bg}	Friction-loss factor
r_{cu}	Copper-loss (resistance) factor
p	Number of pole pairs
K_C	Model scaling or correction factor
$U_{N,\text{verkettet}}$	Rated line-to-line RMS voltage
U_N, I_N, Z_N	Rated phase voltage, rated current, and base impedance
f_N, ω_{EL}	Rated electrical frequency and angular frequency
I_{E0}	Reference excitation current
P_{EL}	Electrical power

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